fn new_conditional_rdp_curve

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March 16, 2026

Proves soundness of `fn new_conditional_rdp_curve` in `mod.rs`.

1 Hoare Triple

Precondition

`base_curve` upper-bounds the base Rényi divergence at every order > 1, and `x` is such that `log_inv_gamma_hi_from_x(x)` succeeds whenever it is called.

Pseudocode

```
def new_conditional_rdp_curve(base_curve, x: float):
    def curve(alpha: float):
        if alpha <= 2.0:  #
            return float("inf")
        eps_alpha = base_curve(alpha)  #  propagates helper errors
        alpha_minus_one_hi = alpha.inf_sub(1.0)  # propagates helper errors
        eps_alpha_minus_one_hi = base_curve(alpha_minus_one_hi)  # propagates helper errors
        alpha_minus_two_hi = alpha.inf_sub(2.0)  # propagates helper errors
        alpha_minus_one_lo = alpha.neg_inf_sub(1.0)  # propagates helper errors
        coeff_hi = alpha_minus_two_hi.inf_div(alpha_minus_one_lo)  # propagates helper errors
        log_inv_gamma_hi = log_inv_gamma_hi_from_x(x)  # propagates helper errors
        term1_hi = coeff_hi.inf_mul(eps_alpha_minus_one_hi)  # propagates helper errors
        term2_num_hi = 2.0.inf_mul(log_inv_gamma_hi)  # propagates helper errors
        term2_hi = term2_num_hi.inf_div(alpha_minus_one_lo)  # propagates helper errors
        return eps_alpha.inf_add(term1_hi).inf_add(term2_hi)  # propagates helper errors
    return curve
```

Postcondition

Theorem 1.1. For every $\alpha > 1$, the returned function either returns `Err(e)` due to a propagated helper error, or returns an upper bound on the Corollary 16 expression

$$\epsilon(\alpha) \;+\; \frac{\alpha-2}{\alpha-1}\epsilon(\alpha-1) \;+\; \frac{2\log(1/\gamma)}{\alpha-1},$$

where $\epsilon(\cdot)$ is the base Rényi divergence curve and $\gamma = 1 - e^{-x}$.

2 Proof

Fix $\alpha > 1$.

Lemma 2.1. If $\alpha \leq 2$, the returned value is $+\infty$, which is a valid upper bound.

Proof. The expression is finite or infinite, and $+\infty$ upper-bounds either case. □

Lemma 2.2. If $\alpha > 2$ and the function returns `Ok(out)`, then `out` upper-bounds the displayed Corollary 16 expression.

Proof. Let $\epsilon(\cdot)$ denote the exact base Rényi divergence curve. By precondition,

$$\texttt{eps_alpha} \geq \epsilon(\alpha).$$

Also $\texttt{alpha_minus_one_hi} \geq \alpha - 1 > 1$. Rényi divergence is nondecreasing in its order, so

$$\epsilon(\alpha - 1) \leq \epsilon(\texttt{alpha_minus_one_hi}) \leq \texttt{eps_alpha_minus_one_hi}.$$

Because $\alpha > 2$, both $\alpha - 2$ and $\alpha - 1$ are positive. The arithmetic helpers therefore give

$$\texttt{coeff_hi} \geq \frac{\alpha - 2}{\alpha - 1}.$$

Since `eps_alpha_minus_one_hi` is nonnegative, upward-rounded multiplication yields

$$\texttt{term1_hi} \geq \frac{\alpha - 2}{\alpha - 1}\epsilon(\alpha - 1).$$

By the proof of `log_inv_gamma_hi_from_x`,

$$\texttt{log_inv_gamma_hi} \geq \log(1/\gamma).$$

Again $\texttt{alpha_minus_one_lo} \leq \alpha - 1$ with a positive denominator, so

$$\texttt{term2_hi} \geq \frac{2\log(1/\gamma)}{\alpha - 1}.$$

Finally, `out` is formed by upward-rounded addition of these three upper bounds, so it upper-bounds the full displayed expression. □

Proof of Theorem 1.1. By Lemma 2.1 for $\alpha \leq 2$, and Lemma 2.2 otherwise. □